

Previous “BK18” Paper (came out fall 2021)

- arxiv.org/2110.00483
- Appendix E.4: Models used
 - 499 Gaussian dust rlz, with $A_d = 3.9 \mu\text{K}^2$
 - Also variants of the Gaussian model (toy decorrelation, amplitude variation)
 - Alternate foreground models with a single realisation, including
 - “PySM 1” = a1d1f1s1
 - “PySM 2” = a2d4f1s3
 - “PySM 3” = a2d7f1s3
 - (see maps [here](#) for those with access)
- Amplitude variation over our field does not lead to bias on r - demonstrated with extreme point source dust model - also see [arxiv/2309.09978](https://arxiv.org/2309.09978)

TABLE II. Uncertainty and bias on r in simulations using a variety of foreground models. For the strongly decorrelated model bias is expected when refit without a decorrelation parameter so this case is in parentheses.

Model	$\overline{A_d}$ (μK^2)	$\overline{A_s}$ (μK^2)	$\sigma(r), \bar{r}/\sigma(r)$			
			no decorr.		with decorr.	
Gaussian	3.9	0.1	0.009,	0.0 σ	0.010,	0.0 σ
G. Decorr.	5.1	0.1	(0.012, +2.1 σ)		0.014, -0.1 σ	
G. amp. mod.	4.4	0.0	0.009,	0.0 σ	0.010,	0.0 σ
PySM 1	11.3	0.9	0.010, +0.1 σ		0.012, +0.2 σ	
PySM 2	25.6	0.8	0.011,	0.0 σ	0.012,	0.0 σ
PySM 3	11.6	0.9	0.011,	0.0 σ	0.013,	-0.1 σ
MHDv3	3.2	7.1	0.012, -0.1 σ		0.013, -0.4 σ	
MKD	3.9	0.1	0.009,	0.1 σ	0.010,	0.0 σ
Vansyngel	5.5	0.1	0.009, -0.1 σ		0.010, 0.0 σ	

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- Also set tight limits on spatial/spectral variation of the dust signal - aka decorrelation as it suppresses cross-spectra relative to auto's
- Decorr parameter likelihood peaks hard at unity - no decorr

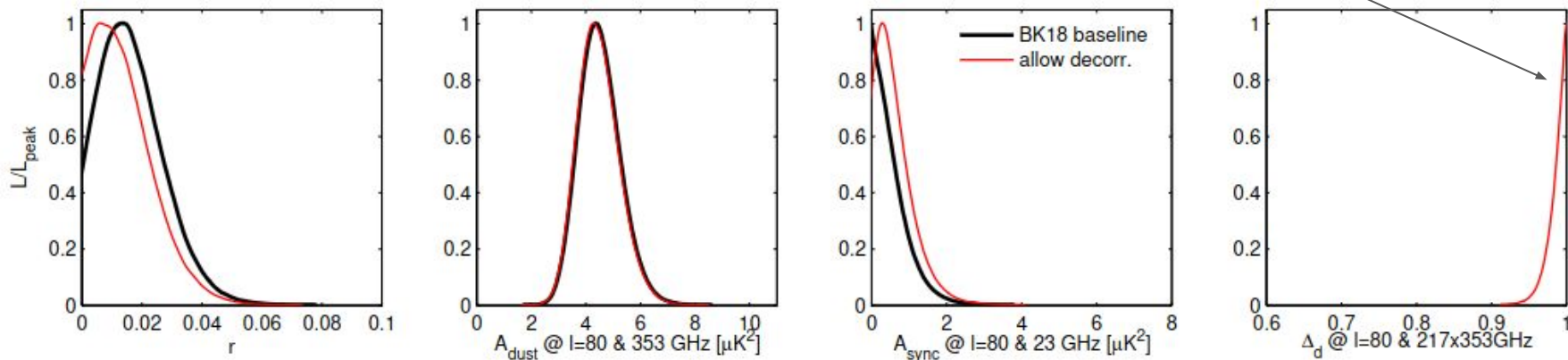


FIG. 18. Likelihood results when freeing dust decorrelation parameter Δ_d . See Appendix E 2 for details.

Latest PySM models vs BICEP data

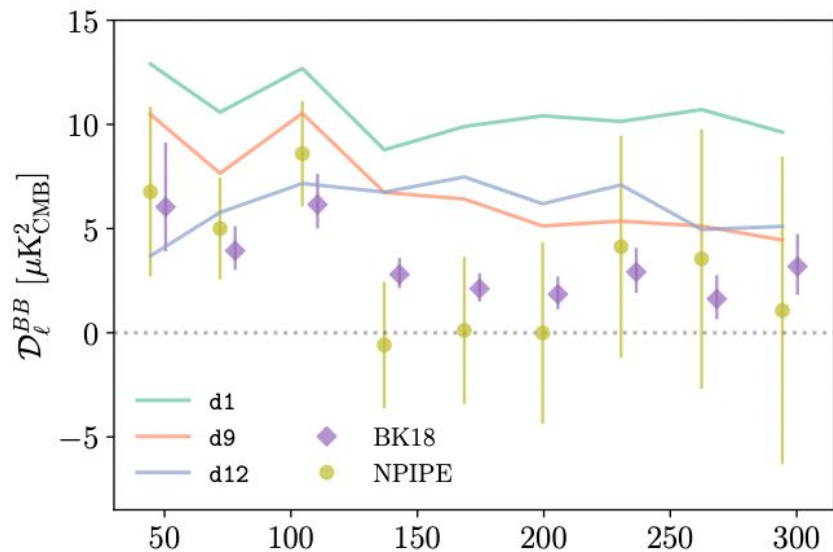


Table 6. Comparison to BK18

	85 GHz	150 GHz	220 GHz	270 GHz	353 GHz
d1	2.42	2.63	2.81	2.94	3.13
d9	2.17	2.12	2.08	2.06	2.03
d10	0.96	1.30	1.59	1.77	2.03
d11	0.97 ± 0.03	1.31 ± 0.04	1.61 ± 0.05	1.80 ± 0.06	2.07 ± 0.07
d12	2.76	2.27	2.08	2.02	1.96

NOTE—Deviation ratios of the reobserved PySM dust BB spectra to the dust component of the BK18 maximum likelihood model at each frequency.

All models overpredict the dust amplitude compared to the observations. BK18 constrains $A_d @ 353\text{GHz} = 4.4 \text{ uK}^2 (+/- 0.8 \text{ uK}^2)$

From PySM paper arxiv.org/2502.20452

Current plans for upcoming BK24 paper

- Dust amplitude now set at the BK18 detection level for Gaussian sims
 - Will include toy dust decorrelation sims like last time
- Planning on including synchrotron at the level we detect (when we do)
 - If no detection, will test inclusion of synchrotron at levels consistent with upper limits
- Will run the latest PySM models... and any others which are available.

What would we like from this group?

- Foreground models to try which are consistent with our data
- Nice if such models are physically well motivated - but consistency with the data is more important...
- E.g. Can we find models of dust spectral complexity (non canonical SED and/or spatial/spectral variation) which are consistent with the data but produce relevant bias on r ?
- We prefer models that make ~ 500 fully independent sky realizations - not locked at low ℓ to the actual sky